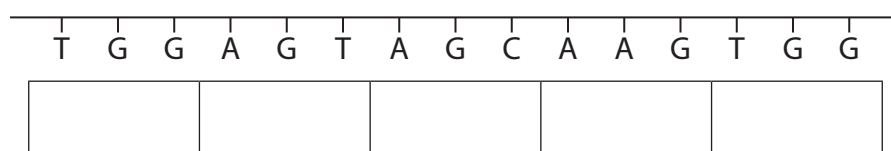


3 The sequence of bases in DNA determines the sequence of amino acids in a polypeptide.

The table shows four amino acids and their genetic codes.

Amino acid	Genetic code
alanine (Ala)	GCT or GCC or GCA or GCG
lysine (Lys)	AAA or AAG
serine (Ser)	AGT or AGC or TCT or TCC or TCA or TCG
tryptophan (Trp)	TGG

(a) The diagram shows a DNA base sequence.



(i) Complete the diagram to show the sequence of amino acids coded by this DNA base sequence.

(1)

(ii) Explain why only five amino acids are coded by this sequence of bases.

(2)

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(b) Explain why some amino acids, such as alanine, have more than one genetic code.

(3)

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(c) Of the 64 possible genetic codes, 61 code for amino acids.

(i) Calculate the percentage of genetic codes that code for amino acids.

Give your answer to four significant figures.

(1)

Answer %



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(ii) Explain the role of the other three genetic codes.

(2)

(d) Messenger RNA (mRNA) and transfer RNA (tRNA) are involved in the synthesis of a polypeptide chain from DNA.

Which row of the table shows the codon on mRNA and the anticodon on tRNA that correspond to tryptophan?

(1)

	Codon on mRNA	Anticodon on tRNA
☐ A	ACC	TGG
☐ B	ACC	UGG
☐ C	UCC	AGG
☐ D	UCC	TGG

(Total for Question 3 = 10 marks)

